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The Book of Evidence

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CHAPTER

12 Evidence for Molecules: Jean Perrin and Molecular Reality

p. 243

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Abstract

Jean Perrin's argument for the existence of molecules on the basis of his 1908 experiments with Brownian motion is examined. Various interpretations of that argument, including hypothetico-deductive, Wesley Salmon's common-cause idea, and Clark Glymour's bootstrapping claims, are examined and rejected. The argument is reconstructed as an eliminative-causal one, and it is shown how it conforms to the requirements of potential evidence. It is also argued, against antirealist interpretations of Perrin, that Perrin himself was applying a realist argument to the existence of unobservable molecules rather than an instrumentalist one to the truth of the observational consequences of the molecular theory.

Keywords: bootstrapping, Brownian motion, common cause, eliminative-causal reasoning, evidence, Glymour, hypothetico-deductive, instrumentalism, Perrin, realism, Salmon

Subject: Philosophy of Science, Epistemology

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The final two chapters of this book invoke case histories in physics: the discovery of the electron, credited to J. J. Thomson in 1897, and Jean Perrin's argument for the existence of molecules in 1908 and 1909 on the basis of experiments involving Brownian motion. Both cases illustrate use of the concepts of evidence I have defined. But each one also involves related philosophical and historical issues I propose to discuss.

In the case of Perrin my philosophical concern is with the question of how observable results Perrin obtained from his experiments could provide evidence that unobservable molecules exist, since the argument he uses seems to presuppose that molecules exist. I will also discuss the historical question of why as late as 1908 Perrin should have thought it necessary to argue that molecules exist. In the case of Thomson my philosophical questions are these: What constitutes a scientific discovery? What sort of epistemic situation does such a discovery require? What sort of evidence? The historical question is whether, in the case of the electron, Thomson was in such an epistemic situation. Did he have such evidence?

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1. Introduction

In 1908 Jean Perrin conducted a series of experiments on Brownian motion from which he drew two conclusions of particular importance: (1) that molecules exist, and (2) that Avogadro's number *N,* the number of molecules in a substance whose weight in grams equals its molecular weight, is approximately 6×10^{23}. Perrin's experimental work and conclusions were set forth in a series of papers published in 1908 and 1909, the most famous of which is his "Brownian Movement and Molecular Reality."[1] An expanded version of his results appeared in his book *Atoms,* published in 1913.[2]

p. 244 In 1926 Perrin received the Nobel Prize in physics primarily for his work on Brownian motion. Despite his considerable success, philosophers and historians of science who read his articles and book should find some of his key arguments puzzling. For one thing, why in 1908, after nineteenth century successes in the kinetic-molecular theory of gases, should Perrin have thought it necessary to argue that molecules exist? Yet argue for this he did.

A second puzzling fact is that Perrin's argument for the reality of molecules seems circular. In brief, from assumptions in kinetic theory involving the existence of molecules, Perrin derives a formula, the "law of atmospheres," that governs a volume of gas. The law relates the number of molecules per unit volume of a gas at a height above some reference plane to Avogadro's number. He then assumes that a slightly modified version of this same law can be applied to the distribution of much larger, microscopic particles (Brownian particles) suspended not in a gas but in a fluid. With this assumption he proposes a formula that relates the number of suspended Brownian particles per unit volume at a height above a reference plane to Avogadro's number and to various experimentally measurable quantities of the visible particles, including their mass, density, and numbers at different heights. He performs experiments measuring these quantities, for different Brownian particles, different liquids, and different temperatures. Each of these measurements, when combined with the law of atmospheres, yields approximately the same value for Avogadro's number. From this fact he concludes that molecules exist. The apparent circularity is that to reach this conclusion Perrin begins by *assuming* that molecules exist. That is an assumption presupposed by the law of atmospheres.

In this chapter I will examine Perrin's reasoning to see whether it is in fact circular. I believe that it is not, and indeed that it conforms with a valid pattern of reasoning frequently used by scientists to infer the existence of "unobservables." I will show how Perrin's experimental results provide veridical (and hence potential) evidence that molecules exist. I will discuss why, even in 1908, it was reasonable for Perrin to employ the pattern of reasoning he does in arguing for the existence of molecules. Finally, I will discuss the relationship between Perrin's reasoning and the debate between realists and antirealists regarding unobservable entities.

2. Perrin's Determination of Avogadro's Number and His Argument for Molecular Reality

Perrin's strategy is first to derive the law of atmospheres for gases.[3] He considers a volume of gas contained in a thin cylinder of unit cross-sectional area and small elevation *h.* The density of molecules making up the gas will p. 245 be greatest at the bottom of the cylinder and decreases exponentially with increasing height. ↳ The pressure p at the bottom of the cylinder is more than the pressure p' at the top (just as air pressure at the bottom of a mountain is greater than at the top). The very small difference in pressure $p - p'$ balances the downward force of gravity gm_c on the mass m_c of gas in the cylinder. So

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$$p - p\prime = \mathrm{gm}_c. \tag{1}$$

Now the mass m_c of the gas is to its volume $1 \times h$ as the gram molecular weight M (the mass in grams equal to the molecular weight of the gas) is to the volume v occupied by a gram molecular weight of the gas. That is,

$$\frac{m_c}{l\,\mathrm{xh}} = \frac{m}{\nu} \tag{2}$$

From (1) and (2) we get

$$p - p\prime = \frac{\mathrm{Mgh}}{\nu} \tag{3}$$

Now Perrin invokes the perfect gas law for one gram molecular weight of a gas

$$\mathrm{pv} = \mathrm{RT}\,,$$

where R is the gas constant and T is absolute temperature, and substitutes $v = RT/p$ for v in (3), obtaining

$$p - p\prime = \frac{\mathrm{Mghp}}{\mathrm{RT}}\,,$$

or

$$\frac{p\prime}{p} = 1 - \frac{\mathrm{Mgh}}{\mathrm{RT}} \tag{4}$$

The pressure of a gas is proportional to its density, and hence to the number of molecules per unit volume. So the ratio p'/p can be replaced by the ratio n'/n, where n' and n are the number of molecules per unit volume at the upper and lower levels, respectively. We obtain

$$\frac{n\prime}{n} = 1 - \frac{\mathrm{Mgh}}{\mathrm{RT}} \tag{5}$$

If m is the mass of a molecule of gas and N is Avogadro's number, then

$$M = \mathrm{Nm}; \tag{6}$$

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that is, the gram molecular weight *M* of a gas is equal to the number of molecules in a gram molecular weight multiplied by the mass of a molecule. So from (5) and (6) we obtain

$$\frac{n\prime}{n} = 1 - \frac{\mathrm{Nmgh}}{\mathrm{RT}} \tag{7}$$

in which Avogadro's number *N* appears.[4]

p. 246 Perrin proposes to use Eq. (7) to determine a value for Avogadro's number experimentally. The problem is that the molecular quantities *n, n′*, and *m* are not directly measurable. So he makes a crucial assumption, viz. that visible particles comprising a dilute emulsion will behave like molecules in a gas with respect to their vertical distribution. In 1827 the English botanist Robert Brown discovered that small, microscopic particles suspended in a liquid do not sink but exhibit rapid, seemingly haphazard motions—so-called Brownian motion. Following Leon Gouy, Perrin assumed that the motions of the visible particles are caused by collisions with the molecules making up the liquid in which the particles are suspended. He also made the assumption that just as invisible molecules that comprise a gas obey the gas laws, so do the visible particles exhibiting Brownian motion in a liquid. Among other things, he assumed that the law (7) derived for molecules in a cylinder of gas could be extended to Brownian particles distributed in a dilute emulsion.

This means that just as the molecules comprising a gas are all identical in mass and volume, so will the Brownian particles have to be. However, in the latter case, the gravitational force acting on a particle will not be its weight *mg,* but its "effective weight," that is, the excess of its weight over the upward thrust caused by the liquid in which it is suspended. This is

$$\mathrm{mg} - \frac{\mathrm{mdg}}{D} = \mathrm{mg}(1 - \frac{d}{D}) \tag{8}$$

where *D* is the density of the material comprising the particles and *d* is the density of the liquid. Replacing the weight *mg* in (7) by the expression in (8), we obtain

$$\frac{n\prime}{n} = 1 - \frac{\mathrm{Nmg}(1 - d/D)h}{\mathrm{RT}} \tag{9}$$

In this equation, *n′* represents the number of Brownian particles per unit volume at the upper level and *n* the same at the lower level; *m* is the mass of a Brownian particle; *N* is Avogadro's number.[5] Equation (9) contains quantities for the suspended particles (not molecules) which Perrin attempted to determine experimentally.

This required the careful preparation of emulsions containing particles equal in size, and determining the
p. 247 density of the material comprising the particles, the ↳ mass of the particles, and (with microscopes) the number of suspended particles per unit volume at various heights—all difficult procedures. Experiments were performed with different emulsions, particles of different size and mass, different liquids, and different temperatures. With various values obtained experimentally for the quantities *n, n′, m, h,* and *T* in Eq. (9), Perrin could use (9) to determine whether Avogadro's number is really a constant, and if so what its value is. He writes:

> In spite of all these variations, the value found for Avogadro's number *N* remains approximately constant, varying irregularly between 65×10^{22} and 72×10^{22} [that is 6.5×10^{23} and 7.2×10^{23}]. (*Atoms,*

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> *p.* 105)

Immediately after this sentence Perrin draws a broader conclusion:

> Even if no other information were available as to the molecular magnitudes, *such constant results would justify the very suggestive hypotheses that have guided us,* and we should certainly accept as extremely probable the values obtained with such concordance for the masses of the molecules and atoms.[6]

Perrin's "suggestive hypotheses" include, of course, the assumption that molecules exist. He continues by noting that the values for Avogadro's number obtained through his experiments agree with the number (6.2×10^{23}) given by kinetic theory from considerations of viscosity of gases. And he concludes:

> *Such decisive agreement can leave no doubt as to the origin of the Brownian movement. . . .* The objective reality of the molecules therefore becomes hard to deny. (*Atoms, p.* 105, italics his)

Perrin's conclusions concerning the value of Avogadro's number and the reality of molecules are drawn from his experiments on Brownian particles suspended in a column of fluid. After drawing them Perrin goes on to consider the theory of Brownian motion developed by Einstein in 1905 which generates an equation relating Avogadro's number to the mean square of the displacement of the Brownian particles in a given direction during a given time. Perrin conducted experiments on such displacement, and using Einstein's equation he generated a value for N close to that achieved by his law-of-atmosphere experiments.[7]

At the end of his book Perrin notes that the value(s) he determined for Avogadro's number approximated ones obtained by a variety of different methods, including ones from experiments on radioactivity, blackbody radiation, and the motions of ions in liquids. And he writes:

p. 248

> Our wonder is aroused at the very remarkable agreement found between values derived from the consideration of such widely different phenomena. Seeing ↳ that not only is the same magnitude obtained by each method when the conditions under which it is applied are varied as much as possible, but that the numbers thus established also agree among themselves, without discrepancy, for all the methods employed, the real existence of the molecule is given a probability bordering on certainty. (*Atoms,* 215–216; see also his 1909 paper, 598–599)

3. Is the Argument for Molecules Circular?

The basic structure of Perrin's reasoning seems to be this.

1. From various assumptions, including that molecules exist, and that gases containing them satisfy the ideal gas law, Perrin derives Eq. (7), which relates the number of molecules at a height h in a container of gas to Avogadro's number and to other quantities including the mass of a gas molecule and the temperature of the gas.

2. Perrin then claims that this formula, or a variation of it, can be applied to visible Brownian particles suspended in a fluid, yielding Eq. (9).

3. Next he devises ways to experimentally measure the quantities in Eq. (9) (other than Avogadro's number), and he conducts various experiments using different fluids and particles.

4. Each of these measurements, when combined with Eq. (9), yields approximately the value 6×10^{23} for N.

5. This approximate value for N is also obtained from experiments other than those involving particles

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suspended in a fluid.

6. From steps 4 and 5 Perrin concludes ("with a probability bordering on certainty") that molecules exist.

The apparent circularity consists in the fact that in step 1 Perrin is making the crucial assumption that molecules exist. Without this assumption he cannot derive Eq. (7), which gives a ratio of the number of molecules per unit volume at the height *h* to the number at the bottom of the cylinder. Is a charge of circularity warranted? In what follows I will consider some attempts to understand Perrin's reasoning so that circularity is avoided.

4. A Common-Cause Interpretation

Wesley Salmon urges that Perrin's reasoning to the reality of molecules is an example of a legitimate common-cause argument. The basic idea of such an argument is this. If very similar effects have been produced, and if it can reasonably be argued that none of these effects causes any of the others, then it can be concluded that these effects all result from a common cause. This, claims Salmon, is how Perrin argues for the reality of molecules: Perrin notes that experiments on various phenomena—including Brownian motion, alpha particle decay, X-ray diffraction, blackbody radiation, and electrochemical phenomena—all yield approximately the same value for Avogadro's number. Salmon asks us to imagine five different scientists engaged in experiments on the five phenomena mentioned. He writes: ↳

p. 249

> These experiments seem on the surface to have nothing to do with one another [so that it is unlikely that one phenomenon studied causes the other]. Nevertheless, we ask each scientist to fill in the blank in this statement: On the basis of my experiments, assuming matter to be composed of molecules, I calculate the number of molecules in a mole [gram molecular weight] of any substance to be ———. When we find that all of them write numbers that, within the accuracy of their experiments, agree with 6×10^{23}, we are as impressed by the "remarkable agreement" as were Perrin and Poincaré. Certainly, these five hypothetical scientists have been counting entities that are objectively real.[8]

Later he says,

> Remember, for instance, the victims of mushroom poisoning; their common illness arose from the fact that each of them consumed food from a common pot. Similarly, I think, the agreement in values arising from different ascertainments of Avogadro's number results from the fact that in each of the physical procedures mentioned, the experimenter was dealing with substances composed of atoms and molecules—in accordance with the theory of the constitution of matter that we have all come to accept. The historical argument that convinced scientists of the reality of atoms and molecules is, I believe, philosophically impeccable.[9]

Since Salmon claims Perrin's argument is "philosophically impeccable," he would deny any circularity charge. According to him, the argument goes like this:

1. If molecules exist, then from experiments on Brownian motion we get a value for Avogadro's number of $N = 6 \times 10^{23}$.

2. If molecules exist, then from Rutherford's experiments on alpha particle decay, we get a similar value for Avogadro's number. The same is true for experiments involving X-ray diffraction, blackbody radiation, and electro-chemical phenomena.

3. There is no reason to suppose that Brownian motion's resulting in a value of $N = 6 \times 10^{23}$ causes alpha particle decay to yield the same value, nor vice versa. The same applies to other cases.

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4. So probably each phenomenon's yielding a similar value for N has a common cause, viz. the existence of molecules.

5. So probably molecules exist.

This argument is not circular, since no assumption is made in premises 1 and 2 that molecules do in fact exist. All that is assumed is a conditional: if molecules exist, then. . . . Salmon himself recognizes this when he writes: p. 250 ↳

> On the basis of my experiments, *assuming matter to be composed of molecules,* I calculate the number of molecules in a mole of any substance to be ———. (p. 221, italics mine)

This corresponds to premises 1 and 2 above. The problem, however, is that this conditional assumption is too weak to yield the strong conclusion 4. The most that premises 1, 2, and 3 warrant is the conditional

4′. *If molecules exist,* then probably each phenomenon's yielding a similar value for N has a common cause.

But 4′ is much less than Perrin himself claims.

To generate the conclusion that Perrin wants, Salmon might alter the argument by adding an additional premise, viz. "molecules exist." But now the argument becomes clearly circular. A more promising approach is to delete the antecedent "if molecules exist" from premises 1 and 2 and assert simply that on each of the varied experiments in question physicists calculated the value of N to be 6×10^{23}, where the latter claim does not presuppose that this is the correct value or even that molecules exist. On this interpretation, we have similar effects (similar calculations of a number that is supposed to represent a number of molecules); and these effects do not cause one another. So by the common-cause principle, we may infer a common cause (without the antecedent assumption that molecules exist). The problem is that a common-cause argument by itself (even assuming its validity)[10] does not permit us to infer what that common cause is but only that there is one. Additional facts must be cited to show that it is the existence of molecules, and not something else, that is the common cause.[11]

One strategy for doing so would be to argue for two points: (a) that the existence of molecules can cause, or be a causal factor in producing, similar calculations of N from experiments on Brownian motion, alpha particle decay, X-ray diffraction, etc. (this could be done by showing how molecular processes can be involved in, or related to, the phenomena in question); and (b) that other possible causes do not produce these effects. Both before and after giving his common-cause argument involving the five hypothetical scientists, Salmon in fact p. 251 goes some way toward arguing for points (a) and (b). He considers how N is related to ↳ the five phenomena cited. And he discusses one rival to molecular theory, viz. energeticism, which, he argues, is incapable of explaining the experimental results. But if indeed it is possible to defend points (a) and (b), then a common-cause argument is both unnecessary and unproductive. It is unnecessary because if (a) and (b) can be successfully defended, then the existence of molecules is shown to be probable without invoking a common-cause argument. It is unproductive because a common-cause argument does not by itself make probable the existence of molecules, contrary to what Salmon claims is shown by his "five hypothetical scientists" argument.

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5. A Hypothetico-Deductive Interpretation

A different interpretation is to suppose that Perrin is engaging in a form of hypothetico-deductive reasoning: From the hypothesis that molecules exist and have properties he attributes to them he draws deductive conclusions regarding observable phenomena, including Brownian motion. He tests these conclusions experimentally and finds they are correct. From this he infers that his molecular hypotheses are probable, or at least that they are confirmed or supported by observations. This is no more circular than any use of hypothetico-deductive reasoning. In its simplest form it is just: *O* is derivable from *T*; *O* is true; hence, *T* is confirmed or probable. *T* is not being assumed to be true or probable at the outset.

From what hypothesis or set of hypotheses is Perrin supposed to have derived observational conclusions, and what observational conclusions does he derive? The following hypothesis is clearly among those from which Perrin derives consequences:

> *h* = Chemical substances are composed of molecules, the number *N* of which in a gram molecular weight of a substance is the same for all substances.

A claim (indeed the most important one) that Perrin establishes experimentally that he takes to confirm *h* is this:

> C. The calculation of *N* done by means of Perrin's experiments on Brownian particles using Eq. (9) is 6×10^{23}, and this number remains constant even when values for *n′*, *n*, and so on, in Eq. (9) are varied.

Proposition *C* might well be called "observational." But it is not something that Perrin derives from his theoretical hypothesis *h*, nor from *h* together with other hypotheses he employs about molecules and Brownian particles. What Perrin does is to derive Eq. (9), not proposition *C*, deductively from such hypotheses. Then he uses Eq. (9) *together with results from various carefully designed experiments*, to establish *C*, which he regards as confirming molecular theory. But this is not the procedure advocated by hypothetico-deductivists. Contrary to the hypothetico-deductive view, the conclusion whose establishment is being claimed to confirm the theory is not derived from that theory.

Even if Perrin does not derive *C* from his theory, could he have done so? Is *C* derivable from the theoretical
p. 252 assumptions Perrin in fact makes? No, because ↳ even though one of the hypotheses Perrin was using is that *N* is a constant, he did not begin with any theoretical postulate concerning the numerical value of this constant. As noted, there were experiments on phenomena other than Brownian motion from which *N* was calculated to be approximately 6×10^{23}. But *C* is not derivable from this fact. Nor, in order to obtain his result *C*, did Perrin assume that these other experimental values for *N* were correct.

Finally, and perhaps most importantly, as I will argue in sections 7 and 8, Perrin's approach to confirming molecular theory is richer than that suggested by a hypothetico-deductive approach. He does not in fact defend this theory simply on the grounds that it entails true "observational" conclusions (whether or not these include *C*). Nor, therefore, is he subject to criticisms of the dubious hypothetico-deductive view of confirmation, according to which if *h* entails *e*, then *e* confirms *h* (see above, chapter 7).

6. Bootstrapping

Clark Glymour's idea of bootstrapping looks more promising than the hypothetico-deductive account because it uses experimental results together with hypotheses in a theory to confirm those hypotheses.[12] To invoke Glymour's own simple example, consider the ideal gas law expressed as

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$$PV = kT\,, \tag{10}$$

where *P* represents the pressure of a gas, *V* its volume, *T* its absolute temperature, and *k* an undetermined constant. We suppose that we can experimentally determine values for *P, V,* and *T,* but not for *k*. The hypothesis (10) can be "boot-strap confirmed" by experimentally obtaining one set of measurements for *P, V,* and *T,* and then employing Eq. (10) itself to compute a value for *k*. Using this value for *k,* together with a second set of values for *P, V,* and *T,* we can instantiate this equation.

Glymour himself cites Perrin's reasoning in determining a value for Avogadro's number as an example of this type of confirmation.[13] Although he does not spell out the Perrin example, presumably what Glymour will say is this: Perrin's Eq. (9) relates Avogadro's number to measurable quantities *n′*, *n, m,* etc. Using one set of measurements for these quantities, Perrin employed Eq. (9) itself to compute a value for *N.* Using this value for *N,* together with a second set of values for *n′*, *n, m,* etc., Perrin instantiated, and thus "bootstrap confirmed," Eq. (9).

In other writings I have criticized Glymour's general account of bootstrap confirmation on the grounds that it allows the confirmation of equations containing completely undefined or obviously meaningless terms.[14] This objection is related to the point I now want to make.

p. 253 When the ideal gas equation *PV* = *kT* gets confirmed in the manner indicated by Glymour—simply by experimentally determining two sets of values for *P, V,* and *T*—the term *k* (at least in Glymour's example) simply represents a constant, that is, a number. This constant can be given a molecular interpretation.[15] But it need not be; it can simply be construed as a constant of proportionality, that is, that number by which *T* needs to be multiplied to yield the same number as the product *PV.* This is the way that Glymour seems to be treating it. The value of that constant is to be determined experimentally.

Now in Perrin's Eq. (9) the constant *N* can be construed in a manner exactly analogous to the way Glymour seems to be treating *k* in (10), that is, as a numerical constant relating the other physical quantities in (9). Indeed, nothing in Glymour's theory of confirmation requires us to interpret *N* in Eq. (9) as a number *of anything,* let alone a number *of molecules.* [16] (The other quantities in (9) are physically interpreted.) Equation (9) would be "bootstrap confirmed" by two sets of measurements of the quantities *n, n′*, etc., if *N* represents the number of angels on the head of a pin, or if *N* is just like a constant of proportionality. So "bootstrap confirming" Eq. (9), or any other equation (such as (7)) containing the constant *N,* does not confirm the existence *of molecules.* (No one, not even Glymour, takes "bootstrap confirming" the ideal gas Eq. (10) to be confirming the existence of molecules, even though *k* in (10) can be given a molecular interpretation.) But when Perrin determined Avogadro's number from his experiments using Eq. (9) he took his results to confirm the existence of molecules. Either he was mistaken in doing so, or else Glymour's "bootstrap confirmation" of Eq. (9) does not capture, at least not completely enough, the logic of Perrin's reasoning.

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p. 254

7. A Solution

I have noted three ways of construing Perrin's reasoning to the reality of molecules from his experimental determination of Avogadro's number on the basis of Brownian motion. Salmon's common-cause idea, as he formulates it, is not sufficient or necessary to yield the desired conclusion. The hypothetico-deductive account does not adequately represent Perrin's reasoning, since his calculation of *N*, from which he infers the existence of molecules, is not derived or derivable from the theoretical assumptions Perrin makes. Nor does Glymour's bootstrapping approach to Eq. (9) permit us to see how Perrin legitimately could have inferred the existence of molecules.

I will now suggest a way to understand Perrin's reasoning that avoids circularity and yields an argument free from problems of the previous interpretations. I will also show how his reasoning yields evidence that satisfies my conditions for potential and veridical evidence.

In section 3 I took the first step in Perrin's reasoning to be (in part) this: From various assumptions, including that molecules exist, and that gases containing them satisfy the ideal gas law, Perrin derives Eq. (7). This certainly is part of Perrin's reasoning, but it is not his "first premise." Both in his 1909 article and in his 1913 book, long before he begins to derive Eqs. (7) and (9), he offers a general discussion of the atomic theory and the existence of atoms and molecules comprising chemical substances. In his book 82 pages are spent developing atomic theory and giving chemical evidence in its favor before he turns to a discussion of Brownian motion in chapter 3 on p. 83. In his article he begins with a description of Brownian motion and then offers arguments that Brownian motion is caused by the agitation of molecules and hence that molecules exist. (This is also how he begins chapter 3 of his book.) Let me briefly mention two such arguments.

First, he writes:

> it was established by the work of M. Gouy (1888), not only that the hypothesis of molecular agitation gave an admissible explanation of the Brownian movement, but that no other cause of the movement could be imagined, which especially increased the significance of the hypothesis.[17]

Perrin notes that Gouy's experiments established that known "external" causes of motion in a fluid, including vibrations transmitted to the fluid by external causes, convection currents, and artificial illumination of the fluid, do not produce the Brownian motion. When each of these known causes was reduced or eliminated the Brownian motion continued unabated. So, Perrin concludes,

> it was difficult not to believe that these [Brownian] particles simply serve to reveal an internal agitation of the fluid, the better the smaller they [the Brownian particles] are, much as a cork follows better than a large ship the movements of the waves of the sea. (p. 511)

Perrin offers a second argument that he regards as stronger than the first. When a fluid is disturbed, the
p. 255 relative motions of its small but visible parts are irregular. ↳ (This can be seen when colored powders are mixed into the fluid.) However, this irregularity of motion does not continue as the visible parts get smaller and smaller. At the level of Brownian motion an equilibrium is established between what Perrin calls "coordination" and "decoordination": if certain Brownian particles stop, then other Brownian particles in other regions assume the speed and direction of the ones that have stopped. From this Perrin draws the following conclusion:

> Since the distribution of motion in a fluid does not progress indefinitely, and is limited by a spontaneous recoordination, it follows that the fluids are themselves composed of granules or *molecules*, which can assume all possible motions relative to one another, but in the interior of which dissemination of motion is impossible. If such molecules had no existence it is not apparent how

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> there would be any limit to the de-coordination of motion. On the contrary if they exist, there would be, unceasingly, partial re-coordination; by the passage of one near another, influencing it (it may be by *impact* or in any other manner), the speeds of these molecules will be continuously modified, in magnitude and direction, and from these same chances it will come about sometimes that neighboring molecules will have concordant motions. (514 –515; italics his)

Perrin is arguing that irregular motions of the parts of a fluid become regular at the level of Brownian particles, in such a way that the total momentum of these particles is conserved. This strongly suggests that the Brownian particles (which are not responsible for their own motion) are being subjected to the influence of smaller particles still—molecules—which exhibit an equilibrium between coordination and decoordination of motion. He concludes:

> In brief the examination of Brownian movement alone suffices to suggest that every fluid is formed of elastic molecules animated by a perpetual motion. (p. 515)

In both arguments Perrin appears to be using eliminative-causal reasoning of the following sort:

(1) A: Given what is known, the possible causes of effect E (for example, Brownian motion) are $C, C_1, \ldots, C_n$ (for example, the motion of molecules, external vibrations, convection currents).

(2) $C_1, \ldots, C_n$ do not produce effect E.

> So probably,
>
> C produces E.

A premise of type (1) may be defended by appeal to the fact that similar known observed effects are produced by and only by one of the types of causes on the list. Alternatively, it may be defended by appeal to more general established principles mandating one of these causes for an effect of that type. (Such principles may also provide a mechanism by means of which E can be produced by one or more of these causes.) A premise of type (2) may be defended by appeal to the fact that effect E is achieved in the absence of $C_1, \ldots, C_n$, or even when these causes are varied. In the first argument, for example, Perrin claims that Gouy's experiments take into account known external causes of motion in a fluid, and that molecular motion can in principle produce
p. 256 Brownian motion, since "the incessant ↳ movements of the [postulated] molecules of the fluid, which striking unceasingly the observed [Brownian] particles, drive about these particles irregularly through the fluid, except in the case where these impacts exactly counterbalance one another" (p. 513). This causal possibility, he obviously believes, can be defended by appeal to more general mechanical principles.[18] In addition, in the first argument Perrin defends premise (2) by appeal to the fact that the motion of the Brownian particles exists whether or not there is external agitation, convection currents, or the like.

Although causation is invoked in these arguments, this is different from Salmon's version of the common-cause argument. For one thing, a common cause of different phenomena is not inferred here. For another, in a common-cause argument no premises of types (1) and (2) need appear. Nor do they in Salmon's "five hypothetical scientists" argument cited earlier. (Nor are they required by the conjunctive fork conditions; see note 11.)

Even if arguments of type A do not establish the existence of molecules with certainty—since other possible causes cannot be precluded with certainty—Perrin believes that his arguments make it likely that Brownian motion is caused by the motion of molecules that make up the fluid. Accordingly, before any discussion of his own experimental results leading to his determination of Avogadro's number, and then to his claim that molecules exist, Perrin presents preliminary reasons to believe the latter claim. (In his article, as well as in his

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book, chemical arguments are also presented, for example, from combinations of elements and compounds). In addition, he presents reasons for believing that Avogadro's number exists, that is, that the number of molecules in a substance whose weight in grams is its molecular weight is the same for all substances.[19] Finally, as Perrin notes, values for Avogadro's number determined by experiments on phenomena other than Brownian motion
p. 257 yield approximately 6×10^{23}.[20] Accordingly, independently of his own experimental results with granules in an ↳ emulsion, Perrin clearly believed that there was information (call it background information *b*) available to him and to other physicists and chemists that supported the following theoretical proposition:

> *T* = chemical substances are composed of molecules, the number *N* of which in a gram molecular weight of any substance is (approximately) 6×10^{23}.[21]

Perrin believed that, on the basis of *b* alone, *T* was at least more probable than not, and hence that

(i) $p(T/b) > \frac{1}{2}$.[22]

He gives arguments (some noted above) to support (i).

Now, as indicated earlier, the experimental result achieved by Perrin on the basis of which he most firmly concludes that molecules exist is

> *C* = The calculation of *N* done by means of Perrin's experiments on Brownian particles using equation (9) is 6×10^{23}, and this number remains constant even when values for *n′*, *n*, etc. in equation (9) are varied.

Proposition *C* is not a deductive consequence of *T*. Even if *T* is true, *C* could be false, since the particular experimental assumptions and conditions introduced by Perrin are not required by *T* to be appropriate to test *T*. To be sure, we might add experimental assumptions to *T* that would yield *C* as a deductive consequence. Such assumptions would include that the Brownian particles of gamboge employed by Perrin all have the same mass and volume, that such particles can be treated like large molecules obeying the standard gas laws, and that Stokes law is applicable to Brownian particles. Perrin gave empirical arguments for each of these and other assumptions he made.[23] But suppose that instead of adding these assumptions to *T*, we simply add to the background information *b* Perrin's experimental results which do not deductively entail these assumptions, but (Perrin believed) made them probable. If so, then, even if *T* together with all the information we are now counting as part of the background information *b* (including other determinations of *N*) does not entail the
p. 258 experimental result *C*, ↳ it does make *C* probable. At least *C* becomes more probable given the truth of *T* than without it. That is, it is more likely that Perrin's experiments will yield $N = 6 \times 10^{23}$ given the assumption that there are molecules whose number $N = 6 \times 10^{23}$, than it is without such an assumption. So,

(ii) $p(C/T\&b) > p(C/b)$.

Now the following is a theorem of probability:

(iii) If $p(C/T\&b) > p(C/b)$, and if $p(T/b) > 0$ and $p(C/b) > 0$, then $p(T/C\&b) > p(T/b)$.[24]

Theorem (iii) states that if *T* increases *C*'s probability on *b*, and if both *T*'s and *C*'s probability on *b* are greater than zero, then *C* increases *T*'s probability on *b*.

Now, from (i), *T*'s probability on *b* is not zero. And, since *b* contains the information that other experimental determinations of *N* yield approximately 6×10^{23}, *C*'s probability on *b* is also not zero. So, it follows from (ii) and (iii) that Perrin's experimental result *C* increases the probability of the theoretical assumption *T*, that is

(iv) $p(T/C\&b) > p(T/b)$.

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Finally, if *T*'s probability on *b* alone is greater than ½, that is, if (i) is true, then we can conclude that *T*'s probability on *C&b* is at least as high, that is,

(v) $p(T/C\&b) > ½$

8. Evidence

Result (iv) will be of interest to those who consider increase in probability as sufficient as well as necessary for evidence. According to this positive relevance view,

(11) *e* is evidence that *h*, given *b*, if and only if $p(h/e\&b) > p(h/b)$.

On this conception, Perrin's experimental result *C* concerning the calculation of Avogadro's number from experiments on Brownian motion counts as evidence in favor of his theoretical assumption *T*, which postulates the existence of molecules. Moreover, the greater the boost in probability that *T* gives to *C* in (ii), the greater the boost in probability that *C* gives to *T* in (iv). Assuming these boosts in probability are high, on the positive relevance conception (11), we can conclude that Perrin's experimental result *C* provides strong evidence for his theoretical assumption *T*. (In section 9 I discuss Perrin's reasons for believing that his evidence was so strong.)

However, as I argue in chapter 4, positive relevance is neither necessary nor sufficient for evidence. In its place I advocate this definition in chapter 8: ↳ p. 259

(PE) *e* is potential evidence that *h*, given *b*, if and only if

(a) *e* and *b* are true;

(b) *e* does not entail *h*;

(c) $p(h/e\&b)\ xp$(there is an explanatory connection between *h* and *e*/*h&e&b*) > ½.

On this conception of evidence, what is important is result (v), not (iv). Increase in probability is neither necessary nor sufficient for evidence. But, it follows from condition (c) in (PE) that high probability is necessary. (c) also requires the high probability of an explanatory connection between *h* and *e*, given the truth of *h&e&b*. But this is also satisfied in the case of Perrin. Given *T*, that chemical substances are composed of molecules, the number *N* of which in a gram molecular weight of a substance is 6×10^{23}, and given *C*, that the calculation of *N* from Perrin's experiments using Eq. (9) is 6×10^{23}, and given the information noted in *b*, the probability is high that the reason *C* obtains is that *T* is true. The probability is high that Perrin's experimental calculation yielded 6×10^{23} because chemical substances are composed of molecules, the number *N* of which in a gram molecular weight is 6×10^{23}. So

(vi) *p*(there is an explanatory connection between *T* and *C*/*T&C&b*) > ½.

If, as seems reasonable, we may also suppose that the probabilities in (v) and (vi) are sufficiently high to allow their product to be greater than ½, then condition (c) in (PE) is satisfied. Assuming that *C* is true, as are the facts reported in *b*, and that *C* does not entail *T*, it follows that Perrin's experimental result *C* is potential evidence for his theoretical claim *T*. Since *T* is true, and since there is an explanatory connection between *T* and *C*, the experimental result *C* is also veridical evidence that *T*.

Perrin's reasoning, so represented, reaches the conclusion that his experimental result from Brownian motion constitutes evidence for the truth of a theoretical claim involving the existence of molecules. It does so on the conceptions of potential and veridical evidence I defend. Moreover, it does so without circularity. There is no undefended assumption at the outset that molecules exist. The claim in (i) that molecules probably exist is

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based on reasons for their existence cited in the background information *b* in (i). Perrin does not begin by assuming without argument that molecules probably exist. He begins by providing a basis for this assumption that includes experiments other than the one of concern to him in *C*. [25]

The pitfalls of the three interpretations in previous sections are avoided. Unlike the common-cause and bootstrap accounts, we end up by confirming a claim entailing that molecules exist. Unlike the hypothetico-deductive account, we need not suppose that Perrin's experimental result expressed in *C* is a deductive consequence of his theoretical assumptions about molecules. Nor need we accept the dubious hypothetico-deductive view of evidence.

p. 260

9. Why Argue for Molecules in 1908?

Suppose we agree that Perrin's reasoning has been adequately represented and that it is not circular. Why in 1908, eleven years after the discovery of the electron, let alone in 1913 when his book appeared, should Perrin have thought it necessary or even useful to present an argument for the existence of molecules?

To begin with, extending into the first decade of the twentieth century, serious opposition to any atomic-molecular theory had been expressed by some physicists and chemists. French and German positivists, including Mach, Duhem, and Poincaré, for whom unobservable entities underlying the observed phenomena were anathema, rejected any realist interpretation of atomic theory. At best, atoms and molecules, if invoked at all, were to be construed simply as instrumental, conceptual devices. The German physical chemist Friedrich Wilhelm Ostwald until at least 1908 rejected atomic theory in favor of the doctrine of "energetics." His grounds for doing so were partly philosophical (a repudiation of any form of unverifiable materialism) and partly based on scientific reasons (including the belief that atomic theories, being purely mechanical, should always entail reversible processes, something incompatible with observed thermodynamic phenomena). Indeed, in the preface to his book *Atoms,* Perrin explicitly mentions Ostwald's rejection of hypotheses about unobserved atomic structure. He describes Ostwald as advocating the "inductive method" which he (Perrin) takes to be concerned only with inferring what is observable from what is observed. By contrast, Perrin is employing what he calls the "intuitive method," which attempts "*to explain the complications of the visible in terms of invisible simplicity*" (*Atoms,* p. vii; his italics).

Although Ostwald rejected atomic theories well into the first decade of the twentieth century, he changed in views by 1909, as a result of the work of Thomson and Perrin.[26] At the end of his 1913 book Perrin refers to the recent controversy over atomic theory and boldly claims that as a result of his (and other) determinations of Avogadro's number,

> the atomic theory has triumphed. Its opponents, which until recently were numerous, have been convinced and have abandoned one after the other the sceptical position that was for a long time legitimate and no doubt useful. (*Atoms,* p. 216)

Perrin was very conscious of the controversies over atoms extending into the twentieth century and felt the need to settle the issue on the side of the atomists.[27]

p. 261

As noted earlier, independently of his 1908 experiments, Perrin believed that available information provided at least some support for the existence of atoms and molecules. Yet he regarded his own experimental results as particularly important in this connection. Why?

Although he considered previous arguments to be supportive, he believed that they did not supply sufficiently direct evidence for molecules. Interestingly, in 1901 he regarded the evidence for the existence of electrons—evidence he himself had helped to develop (see chapter 2)—to be more direct than that for molecules:

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> It is remarkable that the existence of these corpuscles [following J. J. Thomson, Perrin used this term for electrons], thanks to the strong electric charges which they carry, is demonstrated in a more direct manner than that of atoms or molecules, which are much larger.[28]

In 1901, although neither molecules nor electrons were visible as discrete particles observable with a microscope, the effects of electrons were more directly observable than those of molecules. Cathode rays, that is, streams of negatively charged electrons produced in cathode tubes, were observed to produce fluorescence in the glass of the tube, as well as on zinc sulfide screens, and to be deflected by magnetic and electric fields. Neutral molecules and atoms were not known to have these or analogous observable effects. Prior to his experiments, or at least prior to the study of Brownian motion, Perrin regarded the evidence for molecules to be less direct, based as it was in chemistry on the regularities of chemical composition and proportion, and in physics, especially in the kinetic theory of gases developed by Maxwell, on phenomena of heat transfer, on the success of mechanical theories in general, and on the ability of chemical theory as well as the kinetic theory to explain a range of observable phenomena.[29] For Perrin, Brownian motion was for molecules what cathode rays were for electrons. Both phenomena provided a relatively direct link between the postulated entities and their observable effects.

Second, Perrin regarded his evidence for molecules as providing more precise and certain quantitative information about molecules than was previously available. He considered his determination of Avogadro's number and of the masses and diameters of molecules and atoms to be more accurate than previous estimates. He writes:

> this same equation [one corresponding to (9) above] affords a means for determining the constant *N,* and the constants depending on it, which is, it appears, *capable of an unlimited precision.* The preparation of a uniform emulsion and the determination of the magnitudes other than *N* which enter into the equation can in reality be pushed to whatever degree of perfection [is] desired. It is simply a question of patience and time; nothing limits *a priori* the accuracy of the results, and the mass of the atom can be obtained, if desired, with the same precision as the mass of the Earth.[30]

p. 262 Finally, Perrin regarded his experimental results on Brownian motion as important in the confirmation of atomic theory for another reason as well. These results included not only a determination of Avogadro's number from law-of-atmosphere experiments—which has been the focus of attention in this chapter. They also included such a determination from Einstein's theory of Brownian motion, which from kinetic theory assumptions generates a formula relating the displacement of Brownian particles to *N.* Perrin considered Einstein's theory crucial in providing what he called a "mechanism" by which an equilibrium is reached in molecular situations such as those governed by the law of atmospheres. Prior to his 1908 experiments Perrin considered Einstein's theory to be experimentally unverified.

Let us return now to the probabilistic reconstruction of Perrin's reasoning. We formulated his major experimental result as

> C = The calculation of N done by means of Perrin's experiments on Brownian motion using Eq. (9) is 6×10^{23}, and this number remains constant when values for n', n, etc. in Eq. (9) are varied.

A theoretical claim for which C is supposed to provide evidence is

> T = Chemical substances are composed of molecules, the number N of which in a gram molecular weight of any substance is (approximately) 6×10^{23}.

Perrin believed that T's probability is increased by establishing C, that is,

(iv) $p(T/C\&b) > p(T/b)$.

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Indeed, because of the two facts noted above concerning the evidence reported in *C*—its directness and precision—by contrast to other evidence for molecules contained in *b,* Perrin believed that *C* gave a substantial boost to the probability of *T.* If this is right, then for those who adopt the positive relevance account of evidence, and the associated idea that the bigger the increase the stronger the evidence, Perrin's experimental result *C* provided substantial evidence for the theoretical claim *T.* More precisely, on this account of evidence, *C* and *b* together count as stronger evidence for *T* than *b* by itself if and only if (iv) obtains.[31] So on this view of evidence, we can understand at least one reason why Perrin regarded his experimental result *C* as important. Not only did it, together with *b,* provide evidence for *T,* but also it provided stronger evidence for *T* than *b* alone, that is, than information available before Perrin's experiments.

Matters are not so simple on the account of evidence I propose. Although the latter sanctions the conclusion that, given *b,* Perrin's experimental result *C* is potential and veridical evidence for theoretical claim *T,* the question of the strength of that evidence is not settled by (PE). Nevertheless, this much can be said. If, in accordance with (PE), both e_1 and e_2 are potential evidence that *h,* given *b,* and if e_1 reports a higher frequency of the property in question than does e_2, or if e_1 contains a larger, more varied, or more precisely
p. 263 described sample than e_2, or if ↳ it describes items more directly associated with those in *h* than does e_2, then e_1 is stronger evidence for *h* than e_2 *in one or more of these respects.* In Perrin's case, I have argued, definition (PE) is satisfied. His experimental result *C* counts as evidence for *T,* given the background information *b* in question. It can also be argued that *b* itself contains evidence for *T.* But *C* is stronger evidence for *T* than *b* in several respects, including precision and directness. This is among the reasons why Perrin believed his evidential claim was worth making in 1908.

10. Perrin and Realism

Salmon urges that Perrin's argument is an argument for scientific realism. Moreover, unlike the usual philosophical arguments for realism, it is empirical rather than a priori. He writes:

> In an effort to alleviate this intellectual discomfort [produced by philosophical arguments between realists and anti-realists], I decided to try an empirical approach to the philosophical problem. Since it seemed unlikely that scientists would have been moved by the kinds of arguments supplied by philosophers, I felt that some insight might be gained if we were to consider the evidence and arguments that convinced scientists of the reality of unobservable entities.[32]

Salmon believes that Perrin's argument to the reality of molecules provides a "clear and compelling example" of a scientific argument for the existence of unobservable entities. As noted, he considers Perrin's argument to be of the common-cause variety. Even if we reject this interpretation, the question remains as to whether Perrin's argument is, or is best construed as, an argument for scientific realism, and whether Perrin himself understood it in this way.

An antirealist might provide a very different interpretation of Perrin's conclusion. Instead of claiming that the theoretical claim *T* is true, all that Perrin is doing, or at least all that he is entitled to do as a scientist, is infer that *T* is empirically adequate, that it "saves the phenomena." *T* can accomplish the latter without being true. This suggests two questions, one philosophical, one historical. First, is the antirealist correct in supposing that it is possible to have a valid argument to the probable conclusion that some theory saves the phenomena that is not also a valid argument to the probable conclusion that the theory is true? Second, even if it is possible, is it historically plausible to construe Perrin's reasoning in this way?

The answer to the first question is yes, in accordance with the account of "saving the phenomena" provided in chapter 8, section 7. There I showed that it is possible for it to be highly probable that a theory saves the

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phenomena while it is highly improbable that the theory is true.

Applying this to Perrin's reasoning, he conducted a series of experiments on Brownian motion with different values for n', n, and so on, in Eq. (9). Let C_i be the proposition that the calculation of N done by means of
p. 264 Perrin's ith experiment on Brownian particles using Eq. (9) is (approximately) 6×10^{23}. As before, let ↳

> T = Chemical substances are composed of molecules, the number N of which in a gram molecular weight of any substance is (approximately) 6×10^{23}.

In section 7 above, I noted the possibility of taking Perrin's background information b to contain assumptions that together with T entail C.[33] On an antirealist interpretation, Perrin could be arguing that it is highly probable that T saves the C-phenomena without arguing that T is probably true. That is, Perrin could be arguing that

$$(12) \quad \underset{m,n\to\infty}{\text{limit}}\, p(T \text{saves} C_{n+1} \ldots C_{n+m} / C_1 \ldots C_n \& b) = 1$$

without supposing that $p(T/C_1 \ldots C_n \& b)$ is high, or that

$$(13) \quad \underset{n\to\infty}{\text{limit}}\, p(T / C_1 \ldots C_n \& b) = 1.$$

Accordingly, an antirealist has a way of understanding Perrin's reasoning that does not commit the antirealist, or Perrin, to drawing the conclusion that the theory itself is true or highly probable, and hence to drawing the conclusion that molecules are real. As argued in chapter 8, such an antirealist interpretation would preclude our saying that Perrin's experimental results C constitute (potential or veridical) evidence that theory T is true, because the explanatory connection condition would be violated. Nor, I argued, can the antirealist maintain that C constitutes potential or veridical evidence that T *saves the phenomena*—at least not without abandoning antirealism or supplying some yet to be defined concept of evidence that discards an "explanatory connection" condition and avoids the problems that I believe require such a condition.

Even if an antirealist reconstruction is possible—even if all the antirealist wants is to show that Perrin's argument establishes (12) and not that Perrin's experiments provide evidence that molecules exist or evidence that the molecular theory saves the phenomena—is this antirealist reconstruction historically plausible in the case of Perrin?

Admittedly, Perrin makes some remarks which may suggest antirealism. For example, near the end of his paper he writes:

> Lastly, although with the existence of molecules or atoms the various realities of number, mass, or charge, for which we have been able to fix the magnitude, obtrude themselves forcibly, it is manifest that we ought always to be in a position to express all the visible realities without making any appeal to the elements still invisible. But it is very easy to show how this may be done for all the phenomena referred to in the course of this Memoir. (p. 599)

Perrin argues that one can take various laws which relate Avogadro's number to measurable quantities and derive a new equation containing only measurable quantities. If the two laws governing different phenomena are expressible as $N = f(A,B,C)$ and $N = g(D,E,F)$, where A–F are measurable quantities, we can write $f(A,B,C) = g(D,E,F)$, in which (as Perrin puts it) "*only evident realities occur*" (p. 600, his italics). But Perrin does not
p. 265 conclude from this that the most we can ↳ say is that the molecular hypothesis is empirically adequate or has only instrumental value. His main point seems to be that by expressing an equation of the last form above we obtain a result that "expresses a profound connection between two phenomena at first sight completely independent, such as the transmutation of radium and the Brownian movement" (p. 600).

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Most of his comments strongly suggest an attitude of realism. For example, "The real existence of the molecule is given a probability bordering on certainty."[34] Elsewhere, he writes, "Thus the molecular theory of the Brownian motion can be regarded as experimentally established, and, at the same time, *it becomes very difficult to deny the objective reality of molecules.*"[35] These are more typical passages.

Finally, Perrin's argument as I have reconstructed it probabilistically in steps (i)–(v) in section 7 is not an antirealist argument. It is not an argument simply to the conclusion that *T* saves the phenomena, or to (12), a probabilistic version of this. Step (v) asserts the high probability of *T* itself.

Accordingly, an antirealist must show not simply that Perrin's reasoning *can* be reformulated in an antirealist way to the conclusion (12), but that such a reformulation is required or desirable for historical or logical reasons. The historical grounds for such a reformulation are dubious at best. On logical grounds, considering my probabilistic reconstruction, the antirealist would need to show that there are invalid steps in the argument that can be removed only by adopting an antirealist conclusion such as (12) rather than a stronger realist conclusion such as (13). He must show that Perrin's preliminary arguments leading to step (i)—for example, eliminative-causal arguments of type *A* in section 7 from Brownian motion (not appealing to his own experimental results), arguments from chemical combinations, from kinetic theory, and from other determinations of Avogadro's number—do not give *T* a high probability. The antirealist needs to show not simply that these arguments *can be* reformulated as arguments to the conclusion that *T* saves the phenomena, but that something is faulty with these arguments themselves, that they fail to confer high probability on *T* (they fail to establish step (i)). He must show that Perrin's scientific reasoning is erroneous. This is not demonstrated by showing simply that the antirealist conclusion "*T* saves the phenomena" is possible, or even that it is more probable than *T* itself, since it commits one to much less than *T*. In the absence of arguments against specific steps in Perrin's reasoning, one can conclude, with Salmon, that Perrin supplies a reasonable empirical argument for the reality of molecules.[36]

Notes

1. Reprinted in *The Question of the Atom,* ed. Mary Jo Nye (Los Angeles: Tomash Publishers, 1986), 507–601.
2. Jean Perrin, *Atoms* (Woodbridge, CT: Ox Bow Press, 1990; translation of the original *Les Atomes,* 1913, by D. Ll. Hammick).
3. Here I follow the argument in *Atoms,* pp. 90ff; a briefer version is found in "Brownian Movement and Molecular Reality," pp. 529–530.
4. A mathematically more rigorous derivation using differential calculus yields

   $$\frac{n\prime}{n} = \mathrm{e}^{-\mathrm{Nmgh}/\mathrm{RT}}$$

   where *e* is the natural log base. For tiny particles and small *h* the exponent becomes much smaller than 1, and the exponential factor can be expanded in a series whose first two terms are given on the right side of Eq. (7). In his 1909 article, by contrast to his book, Perrin employs the more rigorous derivation.
5. Strictly speaking, in Eq. (9) *N* represents a number for Brownian particles; that is, any quantity of these particles equal to their molecular weight will contain the same number *N* of particles. This number *N* will, according to Perrin's assumptions, be the same as Avogadro's number *N* for molecules.
6. *Atoms, p.* 105, italics his. Using Avogadro's number *N* and the known molecular weights of substances, the mass of molecules is readily determined: mass of a molecule of substance *S* is equal to the molecular weight of *S* divided by Avogadro's number.
7. A useful discussion of these experiments and the statistical reasoning involved is found in Deborah Mayo, "Cartwright, Causality, and Coincidence," *PSA,* 1986, vol. 1, 42–58, and Mayo, "Brownian Motion and the Appraisal of Theories," in A. Donavan et al., eds., *Scrutinizing Science* (Dordrecht: Kluwer, 1988), 219–243 10.1007/978-94-009-2855-8_11 .
8. Wesley Salmon, *Scientific Explanation and the Causal Structure of the World* (Princeton: Princeton University Press, 1984), p. 221.
9. *Ibid.*, p. 223. Nancy Cartwright, *How the Laws of Physics Lie* (Oxford: Oxford University Press, 1983), 82–85, offers a

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somewhat similar analysis of Perrin's reasoning. She takes it to be an argument to the "most probable cause." Experiments on seemingly unrelated phenomena all yield the same calculation for Avogadro's number. "Would it not be a coincidence if each of the observations was an artefact, and yet all agreed so closely about Avogadro's number? The convergence of results provides reason for thinking that the various models used in Perrin's diverse calculations were each good enough. It thus reassures us that those models can legitimately be used to infer the nature of the cause from the character of the effects" (pp. 84–85). For a criticism of Cartwright's account, see Deborah Mayo, "Cartwright, Causality, and Coincidence," *PSA,* vol. 1, pp. 42–58.

10. For a general criticism of common cause arguments, see Bas van Fraassen, *The Scientific Image* (Oxford: Oxford University Press, 1980) 10.1093/0198244274.001.0001 , and Frank Arntzenius, "The Common Cause Principle," *PSA,* 1992, vol. 2, 227–237.
11. In explicating the idea of a common cause, Salmon employs Reichenbach's notion of a conjunctive fork defined probabilistically in terms of these 4 conditions: (i) $p(A\&B/C) = p(A/C)p(B/C)$; (*ii*) $p(A\&B/-C) = p(A/-C) = p(B/-C)$; (*iii*) $p(A/C) > p(A/-C)$; (*iv*) $p(B/C) > P(B/-C)$. These conditions are satisfied, Salmon argues, if *A* and *B* represent experimental results from two different phenomena yielding the same value for Avogadro's number, and *C* includes the assumption that molecules exist. But as Salmon himself recognizes (pp. 167–168), these are not sufficient conditions for *C* to be a common cause of *A* and *B.* Nor, indeed, does the satisfaction of these conditions make it highly probable that *C* is a common cause of *A* and *B.* Incompatible *C*'s could satisfy these conditions. (For an example, see my coin-tossing case, in chapter 6, section 10.) Yet Perrin (as well as Salmon) wants to conclude that, in all probability ("bordering on certainty"), molecules exist. The satisfaction of the conjunctive fork conditions will not yield such a conclusion.
12. Clark Glymour, *Theory and Evidence.*
13. Glymour, "Relevant Evidence," reprinted in Peter Achinstein, *The Concept of Evidence* (Oxford: Oxford University Press, 1983), p. 130, fn. 12.
14. Peter Achinstein, *The Nature of Explanation,* chapter 11; see note 16 below.
15. When Glymour writes Eq. (10) he does not make clear which constant he has in mind by *k.* Perhaps he means what physicists usually call *R*, the universal gas constant, which is experimentally determined in the manner he notes. Perhaps he means Boltzmann's constant *k,* which is equal to *R* divided by Avogadro's number *N.* In either case, Glymour's *k* in Eq. (10) can be related to molecular quantities.
16. It is instructive to look at an example which Glymour does work out involving 6 equations: (1) $A_1 = E_1$; (2) $B_1 = G_1 + G_2 + E_2$; (3) $A_2 = E_1 + E_2$; (4) $B_2 = G_1 + G_2$; (5) $A_3 = G_1 + E_1$; (6) $B_3 = G_2 + E_2$. The *A*s and *B*s are directly measurable quantities, the *E*s and *G*s are "theoretical" quantities whose values can be determined indirectly through the theory by determining the values of the *A*s and *B*s. (Glymour identifies none of these quantities.) According to Glymour, here is how we can confirm hypothesis (1). We determine a value for A_1 directly by experiment, since this quantity is directly measurable. We obtain a value for the theoretical quantity E_1 by obtaining values for the observables B_1, B_3, and A_3, and then by using hypotheses (2), (5), and (6), mathematically computing a value for E_1. If this value is the same as the one determined for A_1 directly by experiment, then we have confirmed hypothesis (1). Note that we can confirm (1) without assigning any physical meaning to E_1. Only the observable quantities (the *A*s and *B*s) need be given any physical meaning.
    In a simple counterexample to Glymour's system I develop in *The Nature of Explanation,* chapter 11, let *A* = the total force acting on a particle; *B* = the product of the particle's mass and acceleration; *C* = the quantity of God's attention focused on a particle. The theory consists of two equations: (i) *A* = *C*; (ii) *B* = *C*. On Glymour's account, these equations can be bootstrap-confirmed by measuring *A* and *B* and using equations (i) and (ii) in the manner Glymour proposes. Surely one cannot conclude that anything about God has been confirmed by such a procedure!
17. "Brownian Movement and Molecular Reality," pp. 510–511.
18. At this point, however, Perrin does not present such principles or show quantitatively exactly how molecular motion can cause Brownian motion. This is important because some earlier investigators claimed that Brownian motion was incompatible with standard assumptions of the kinetic-molecular theory. One objection, first raised by Karl Nägeli in 1879, is that kinetic theory calculations show that the velocity that would be imparted to a Brownian particle by a collision with molecules would be much too small to observe, contrary to what is the case. A second objection is that Brownian motion, if produced by molecular collisions, would violate the second law of thermodynamics. (For a discussion of the validity and impact of these objections, see Mary Jo Nye, *Molecular Reality: A Perspective on the Scientific Work of Jean Perrin* (London: Macdonald, 1972), pp. 25–27, 101–102; Stephen G. Brush, "A History of Random Processes," *Archive for History of the Exact Sciences,* 5, 1968, pp. 10ff; Roberto Maiocchi, "The Case of Brownian Motion," *British Journal for the History of Science,* 23 (1990), pp. 261ff.
    Although Perrin in the qualitative discussion of Guoy does not present a quantitative mechanical explanation of Brownian motion, he does later in his article when he gives the theoretical explanation offered by Einstein in 1905. He responds to the second objection by defending a statistical, by contrast to a universal, interpretation of the second law of

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thermodynamics—one that he regards as established by Clausius, Maxwell, Helmholtz, Boltzmann, and Gibbs (see Perrin 1984, p. 512).

19. See "Brownian Movement and Molecular Reality," 515–516; *Atoms,* 18ff.
20. *Atoms,* 105, 215; "Brownian Movement . . . ," 521–524, 583–598.
21. Note that *T,* unlike *h* in section 5, gives a specific value for Avogadro's number. (*h* is simply "Chemical substances are composed of molecules, the number *N* of which in a gram molecular weight of a substance is the same for all substances.") The point of the difference is this. Perrin's derivation of Eq. (7), and then of Eq. (9), proceeds from *h,* not from the stronger *T* (which contains a specific value for *N*). Otherwise in Eqs. (7) and (9) he could have substituted a specific value for *N.* So from the perspective of the h-d account, which considers what theoretical hypotheses Perrin starts with, from which he derives conclusions (7) and (9), it is appropriate to choose *h* rather than *T.* However, my point in the present section is not to consider Perrin's initial theoretical hypotheses, but rather hypotheses he takes to be proved or made likely by his arguments. Clearly the stronger hypothesis *T* (which entails the weaker *h*) is what Perrin seeks to establish.
22. Probability is here construed in the objective epistemic sense of chapter 5.
23. For example, in "Stokes' Law and Brownian Motion," *Comptes Rendus* (1908), 147ff, he gives experimental arguments for the applicability of Stokes' law. For a discussion and criticism of these arguments, see Roberto Maiocchi, "The Case of Brownian Motion," 257–283; see 278–279.
24. Proof: According to Bayes' theorem, $p(T/C\&b) = p(T/b) \times p(C/T\&b)/p(C/b)$. By the assumptions of the theorem, $p(T/b) > 0$ and $p(C/b) > 0$. So $p(T/C\&b)/p(T/b) = p(C/T\&b)/p(C/b)$. Since, by the assumptions of the theorem, $p(C/T\&b) > p(C/b)$, it follows that $p(T/C\&b) > p(T/b)$.
25. As indicated in note 18, it also includes the theoretical assumption, later justified, that a quantitative mechanical explanation of Brownian motion can in principle be obtained from kinetic theory assumptions.
26. "*I have convinced myself that we have recently come into possession of experimental proof of the discrete or grainy nature of matter, for which the atomic hypothesis had vainly been sought for centuries, even millenia.* The isolation and counting of gas ions on the one hand . . . and the agreement of Brownian movements with the predictions of the kinetic hypothesis on the other hand, which has been shown by a series of researchers, most completely by J. Perrin—this evidence now justifies even the most cautious scientist in speaking of the *experimental* proof of the atomistic nature of space-filling matter" (Friedrich Wilhelm Ostwald, *Grundriff der Allgemeinen Chemie* (Leipzig: Engelmann, 1909; quoted in Brush, "A History of Random Processes,"; italics Ostwald's).
27. For a very informative extended discussion, see Mary Jo Nye, *Molecular Reality: A Perspective on the Scientific Work of Jean Perrin* (London: Macdonald, 1972).
28. Quoted in Nye, *Molecular Reality,* 83–84; from "Les Hypotheses Moleculaires," *Revue Scientifique,* 15 (1901), 449–461; quotation, *p.* 460.
29. For Maxwell's arguments for molecules, see my *Particles and Waves,* essays 7 and 8.
30. "Brownian Movement . . . ," 555–556.
31. On a standard view, e_1 provides stronger evidence than e_2 for *h* if and only if $p(h/e_1)-p(h) > p(h/e_2)-p(h)$, that is, if and only if $p(h/e_1) > p(h/e_2)$. See chapter 3.
32. Salmon, *Scientific Explanation* . . . , 213–214.
33. Although this possibility was noted, in that section I supposed only that *T* increases *C*'s probability on *b.*
34. *Atoms,* 215–216.
35. "Brownian Movement . . . ," 554, italics his.
36. For very helpful criticisms of material in this chapter, I am indebted to Laura J. Snyder, Robert Rynasiewicz, Nancy Cartwright, Michael Redhead, students in my 1994 graduate seminar at Johns Hopkins, and four excellent referees for *Perspectives on Science* in which a version of this chapter initially appeared.

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